United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam, Trimester: Fall 2023

Course Code: CSE 6093

Course Title: Research Methodologies and Technical Writing

Total Marks: 40 Duration: 2 hours Section: M & N

There are total 5 questions. Answer all questions.

1. (a) Paper: Digital health inclusion towards achieving universal health coverage for Bangladesh utilizing general practitioner model. **2**

Question: State the goal of this research paper in one sentence. What objectives author want to achieve here? How many participants used in this research?

Answer:

- ✓ **The goal of this research paper:** Necessary steps should be taken to raise preventive and primary healthcare awareness about non- communicable diseases among rural people.
- ✓ **Objective:** Bangladesh's health care system, particularly in rural areas, experiences enormous obstacles in providing complete preventive and primary healthcare services due to the lack of adequate healthcare facilities, resource constraints, and a non-functional referral system. To alleviate these problems, in this study, we introduce the digital general practitioner (GP) model for rural Bangladesh, digital platforms and present a statistical analysis of the data that was gathered from the pilot project.
- ✓ 12,746 participants have been used in this research.

(b) Paper: Swallowing accelerometry signal feature variations with sensor displacement. **2**

Question: State the goal of this research in one sentence. How many feature domains they used, and list all features name? Which statistical method used to test significant difference among position variations and also why used this method?

Answer:

- ✓ **The goal of this research paper:** Routine monitoring and improving of swallowing function for neurological disorders patients such as stroke, cerebral palsy, and Parkinson's disease.
- ✓ Three feature domains have been used. There are Time domain features, Frequency domain features and Time-Frequency domain features.
- ✓ Due to the non-normality of the feature value distributions, the non-parametric Friedman test was employed with $\alpha = 0.05$. Post-hoc analyses for features with significant position effects were performed using Wilcoxon signed rank tests with $\alpha = 0.05$.

(c) Paper: Development of an Intelligent Platform to Digitize Marriage, Empower Married Couple for Life and Family Wellbeing Towards Digital Bangladesh. **2**

Question: State the goal of this research in one sentence. What objectives author want to achieve here? Write the outcome of this research?

Answer:

- ✓ **The goal of this research paper:** The aim of this paper is to analyze the primitive way marriage system and their huge impacts and effects on social life, finding the solution and generating a way to stop the delusion of the Kazi.
- ✓ **Objectives:** There are two separate phases to cover the full objective of this paper- First one is Digitization in marriage registration system and second one Empowerment of community health service for a married couple into their conjugal life.
- ✓ **Outcome:**
  - In this article examined the problems concerning the primitive way of marriage registration, married couples, pregnant women, baby nutrition and health.
  - Proposed digitization of marriage system will reduce the early marriage rate.
  - The awareness will be created among the general mass.

(d) Paper: CMED: Cloud based medical system framework for rural health monitoring in developing countries. **2**

Question: State the goal of this research in one sentence. Is this research having any social impact? If yes, please write it.

Answer:

- ✓ **The goal of this research paper:** Primary prevention intervention model to prevent the exposure to risk factors of NCDs remote areas in Bangladesh.
- ✓ **Social impact:** CMED Health is bringing upon change and creating impact in the health sector by saving lives. Regarding feasibility we have identified the following advantages of CMED digital healthcare platform.
  - Through this smart app health managers can constantly supervise the health workers and easily identify the gap in the system.
  - Due to the collection and storage of information in the digital system the usability of the app has been increased and now it is possible to do the research in various directions like socio-economic, gender based, environmental, disease risk factors identification and association with different factors.
  - Health administrators and policy makers can visualize the overall information of a population in a defined geographical area sitting in one place through the dashboard of the platform, like the total birth, death, marriage, pregnancy, disease prevalence, sufferings, medical expenses etc.
  - Collecting and recording the information at the same time has reduced the additional hassle for further data collection.
  - As the digital system works in offline, so the health workers could work uninterruptedly and finish their work in time even though the areas don’t have good internet coverage.

- Using this app, the workload of the health workers has been reduced a lot and their interest in work has been increased. As a result, the efficiency of the organization is also increasing.
- The knowledge of the community people regarding their health risks and its prevention has been increased as instant advice are provided through the CDSS of the app during health monitoring.

2. (a) Illustrate the process of identifying a research problem and developing research questions. **2**

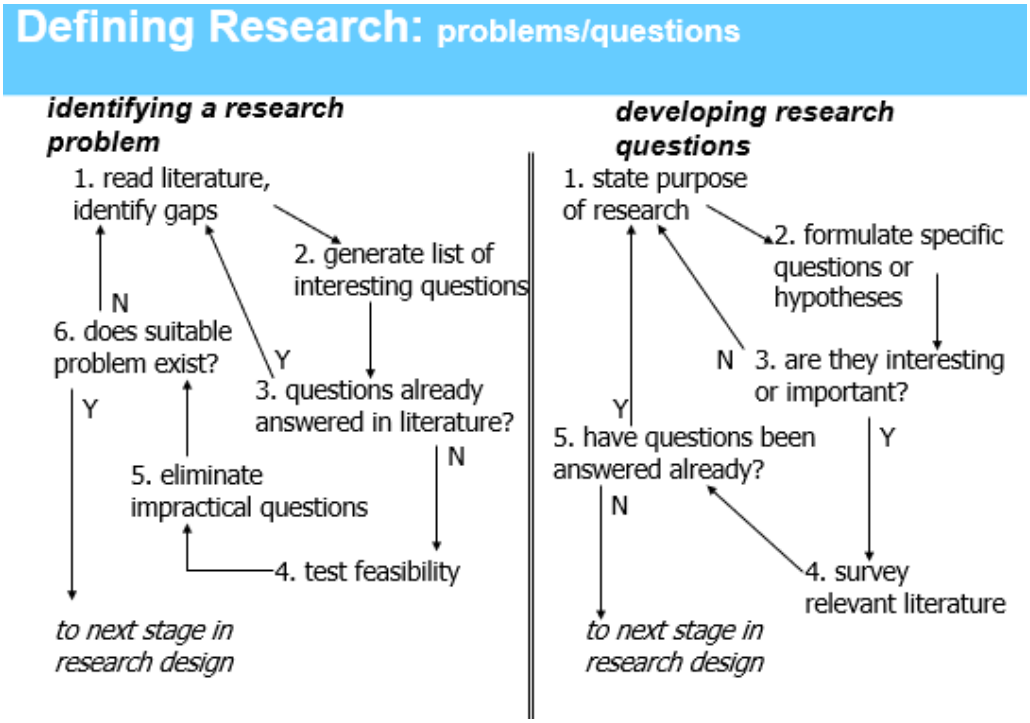
Ans:

Identifying a research problem

i. Read literature, identify gaps.
ii. Generate list of interesting questions.
iii. Questions already answered in literature?
iv. Test feasibility.
v. Eliminate impractical questions.
vi. Does suitable problem exist?
vii. To next stage in research design.

Developing research questions

i. State purpose of research
ii. Formulate specific questions or hypotheses.
iii. are they interesting or important?
iv. Survey relevant literature.
v. Have questions been answered already?
vi. To next stage in research design



(b) Dhaka north city corporation (DNCC) is facing problem to manage garbage collection and disposal in the city. They are looking for smart solution for it and want to make the city with a vision, "Clean City is Smart City" to improve the situation and create awareness. Your task is to propose a solution based on your research knowledge that can address the problem and help to achieve the vision. **2**

To help Dhaka North City Corporation (DNCC) achieve its vision of "Clean City is Smart City," I propose a **Smart Waste Management System** based on the integration of IoT (Internet of Things) technology, data analytics, and citizen engagement. Here's a solution that can address the problem and improve garbage collection and disposal:

1. IoT-Enabled Smart Bins

- **Smart Garbage Bins** equipped with sensors to monitor fill levels in real-time. These sensors can detect when bins are nearly full and send automatic notifications to the waste management authorities.
- **Data Transmission**: The data collected by the sensors is transmitted to a central system via a wireless network, enabling real-time monitoring and optimized waste collection routes.

2. Optimized Waste Collection Routes

- **Data Analytics**: The data from smart bins can be analyzed to generate optimized routes for garbage collection trucks, reducing unnecessary trips and fuel consumption.
- **Dynamic Scheduling**: Waste collection can be dynamically scheduled based on real-time bin status, ensuring only full bins are collected, thereby minimizing operational costs and improving efficiency.

3. Citizen Engagement and Awareness via Mobile App

- **Mobile Application**: A mobile app can be developed to engage citizens in waste management. Features can include:
    - **Real-time Notifications**: Inform citizens of nearby full bins and direct them to the closest empty bin.
    - **Report Garbage Issues**: Allow citizens to report garbage overflow, illegal dumping, or waste-related issues.
    - **Awareness Campaigns**: Conduct awareness programs on waste reduction, recycling, and proper garbage disposal through the app.

4. Waste Segregation and Recycling

- **Color-Coded Bins**: Implement a smart bin system that supports waste segregation at the source (e.g., bins for recyclables, organic waste, and general waste).
- **Recycling Incentives**: Introduce reward points for citizens who correctly dispose of waste using the app. Points can be redeemed for discounts on utility bills or public transport passes, encouraging participation.

5. Centralized Monitoring System

- A **centralized waste management dashboard** for city officials to monitor garbage levels, optimize resource allocation, and track collection truck movement. This system would help in identifying problem areas, scheduling maintenance, and assessing the overall efficiency of waste management operations.

6. Public Awareness and Education

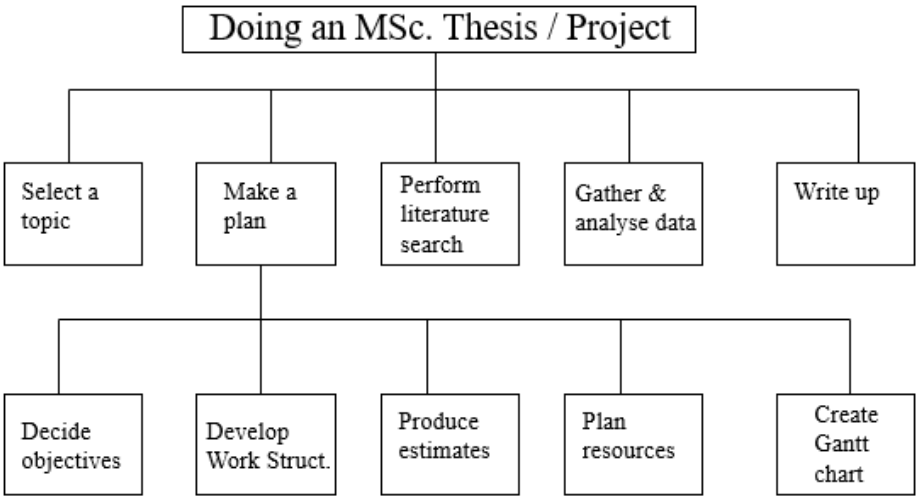
- Launch a **"Clean City is Smart City" Campaign** through social media, mobile apps, and local events to educate citizens about their role in waste management and the use of smart bins and recycling initiatives.

7. Sustainable Waste-to-Energy Solutions

- **Waste Processing Facilities**: Integrate waste-to-energy plants that convert non-recyclable waste into energy. This can reduce landfill waste and contribute to the city's energy supply, creating a circular waste management system.

(c) List the basic components of writing a thesis or research paper. **2**

L2



(d) Why we need literature survey for research design? Write the structure of writing literature survey in the research paper. **2**

Need literature survey: A literature survey is essential for research design because it helps identify existing knowledge, gaps, and trends in the field. It ensures that the research is built on a solid foundation and avoids duplication of previous work. Additionally, it guides the formulation of research questions and methodologies.

L3

i. knowledge of current literature and your own reading.

ii. present it thematically or in an order that relates to the rest of the project structure.

iii. identify trends, gaps etc. in the literature.

iv. analyses, evaluate and otherwise demonstrate that you have understood what you have read.

Structuring the literature survey

- Section 2.1 (introduction)
  - Introduce your own research topic and explain what literature areas need to be reviewed.

- Briefly present your concept map or plan to show the reader how the literature review is logically structured.

- Section 2.2 to 2.x
  - Address each of the main areas of the concept map or plan.
  - Try to move towards specific discussion of each area and identify what is already known and what is not known.

- Section 2.x (final section)
  - End by stating your own research challenge/question or any hypothesis.

3. (a) Distinguish between null hypothesis and hypothesis. **1**

-Null hypothesis is usually a statement of "no effect" or "no difference" between variables.

-A hypothesis is a general statement or prediction about of "causal" relationships between dependent and independent variables.

In essence:

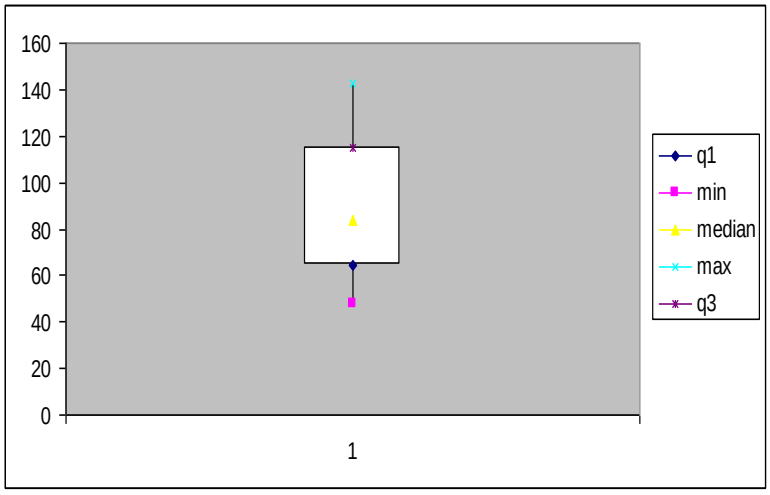
- **Hypothesis**: There **is** a relationship or effect.
- **Null hypothesis**: There is **no** relationship or effect.

(b) When we use Box Plot. Illustrate Box Plot with an example. **3**

Use case: We use a box plot to visually represent the distribution, variability, and outliers in a dataset, particularly when comparing the central tendency and spread across different groups or variables.

Box Plot: A box plot is a graph of the five number summary. The central box spans the quartiles. A line within the box marks the median. Lines extending above and below the box mark the smallest and the largest observations (i.e., the range). Outlying samples may be additionally plotted outside the range.

Distribution of Age in Month



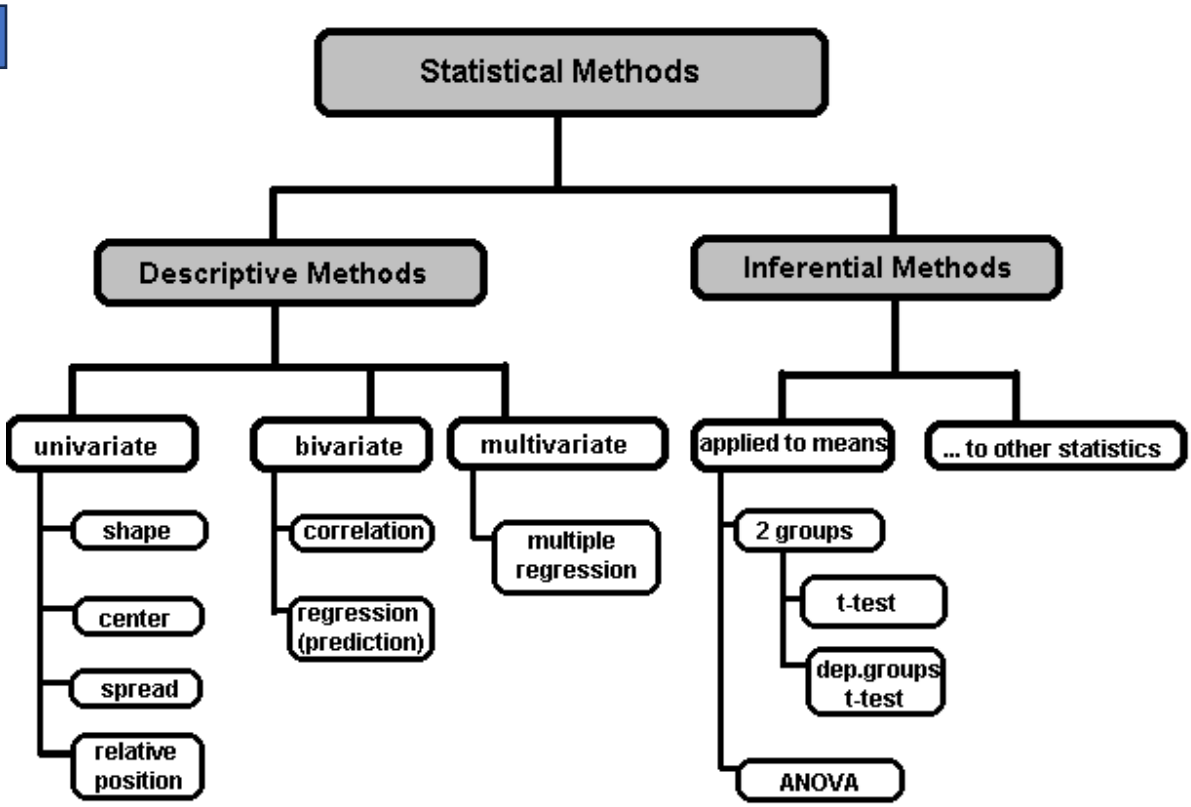
(c) Write the difference between independent and a dependent variable(s). **1**

An **independent** variable is the factor that is manipulated or controlled in an experiment to observe its effect, while a **dependent** variable is the outcome or response that is measured and expected to change based on variations in the independent variable.

(d) Show the taxonomy of Statistics for data analysis. Or **2**

List different statistical tests that we use for evaluating research hypothesis.

L8



(e) Why we need to do statistical "significance" tests in our research? **1**

-We use of statistical “significance” tests to prove / disprove hypothesis.

-Tests provide “court-room” evidence that our hypothesis is true or false.

4. (a) What is data distribution? Why it is import in the quantitative analysis. List different data distributions. **2**

Data distribution refers to the way data points are spread or arranged in a dataset. It shows how often different values occur, providing insights into the shape, spread, and central tendency of the data. Data distribution can be visualized using graphs like histograms or probability density functions.

Important in Quantitative Analysis

Data distribution is crucial in quantitative analysis because:

1. **Informs the choice of statistical tests**: Different tests assume specific distributions (e.g., normal distribution for parametric tests).
2. **Helps identify patterns**: Understanding the distribution helps in identifying trends, outliers, and data characteristics.
3. **Guides decision-making**: The shape of the data distribution influences key conclusions and predictions.

List of Different Data Distributions:

1. **Unimodal** - having a single peak.
2. **Bimodal** - having two distinct peaks.
3. **Symmetric** - left and right half are mirror images.

Each of these distributions has unique characteristics and is used for specific types of data analysis.

(b) Describe the difference between Qualitative and Quantitative analysis for research evaluation. **2**

Qualitative	Quantitative
(Usually) Non-probability based sample	Typically a probability-based sample
Non-generalizable	Generalizable
Answers Why? How?	Answers How many? When? Where?
Formative, earlier phases	Tests hypotheses, latter phases
Data are "rich" and time-consuming to analyze	Data are more efficient, but may miss contextual detail
Design may emerge as study unfolds	Design decided in advance
Researcher IS the instrument	Various tools, instruments employed

Qualitative Analysis

i. Non numerical
ii. Case studies
iii. Action research
iv. Thought experiments.

Quantitative Analysis

i. Numerical
ii. Experiments and surveys with numerical data
iii. Statistical techniques used to prove / disprove hypothesis.

(c) Illustrate Mean, median, interquartile range, coefficient of variation with example. 4

Mean: Summing up all the observation and dividing by number of observations. Mean of 20, 30, 40 is (20+30+40)/3 = 30.

Median: The middle value in an ordered sequence of observations.

For example, to find the median of 3, 5, 6, 7, 9 and the middle value 6 this is median.

The median is less sensitive to outliers (extreme scores) than the mean and thus a better measure than the mean for highly skewed distributions, e.g. family income. For example mean of 20, 30, 40, and 990 is (20+30+40+990)/4 =270. The median of these four observations is (30+40)/2 =35. Here 3 observations out of 4 lie between 20-40. So, the mean 270 really fails to give a realistic picture of the major part of the data. It is influenced by extreme value 990.

Mode: The value that is observed most frequently.

Interquartile range: In statistics, the **interquartile range (IQR)** is a measure of statistical dispersion that represents the range within which the middle 50% of the data falls. It is calculated by subtracting the first quartile (Q1), which is the 25th percentile, from the third quartile (Q3), which is the 75th percentile.

Q1: The value below which 25% of the data lies.
Q3: The value below which 75% of the data lies.

Coefficient of Variation: The standard deviation of data divided by it's mean. It is usually expressed in percent.

Coefficient of Variation = $\frac{\sigma}{\bar{x}} \times 100$

5. (a) Illustrate the principle of Actors and Actions in Scientific Writing. **2**

Ans:

Principles:

1. **Put actions in verbs.** Scientific writers often use "clunky" (awkward) abstract nouns instead of "spunky" (strong, interesting) verbs.
2. **Put actors in subjects.** Scientific writers often use abstract nouns to convey main action, use weak verbs.
3. **Keep subjects near verbs.**

(b) Why Cohesion, Coherence, and Emphasis is important in writing a research article. 2

How can we ensure that in our writing?

Cohesion, **coherence**, and **emphasis** are crucial elements in writing a research article because they ensure clarity, logical flow, and the effective communication of key ideas.

Principles:

i. Put new information last (cohesion)
ii. Use passive voice judiciously.
iii. Make sure the first and last sentences of a paragraph match (coherence)

Here's why they are important:

Cohesion: Refers to how well the different parts of the text connect through the use of linking words and phrases.

Coherence: Involves the logical structure and flow of ideas throughout the article.

Emphasis: Ensures that the most important points in the research article stand out. It helps guide the reader's attention to critical findings, arguments, or conclusions by using techniques such as topic sentences, repetition, or strategic placement of key information.

(e) As we know simplicity increase readability, so how can we improve simplicity by reducing complexity is our writing? 2

Principles:

i. **Omit needless words (excessive hedging, ineffectual phrases).** For example, Multiword phrases that mean nothing beyond a single word. Instead of "**prior to",** Consider "**before".**
ii. **Prefer simple words.** For example, never use a complex word when a simple one will do. Instead of "**utilize",** Consider "**use".**
iii. **Use simple subjects.** Scientific writing abounds with complex sentence subjects.
iv. **Use adjectives/adverbs frugally.** Often adjectives and adverb modifiers add no meaning.

(d) Consider following sentence. Is this sentence express the writers view easily? If not then rewrite it in a way that reader can get respective information easily. 2

To assess the degree of enrichment of enhancer activities in predicted tissues, we compared the relative frequency of enhancers for each of the three tissues examined here with a background set of 528 previously tested sequences predicted to be developmental enhancers on the basis of extreme sequence constraint that were not associated with a prior tissue specificity prediction.

Ans: To assess enhancer activity in predicted tissues, we compared the frequency of enhancers in the three tissues with a background set of 528 sequences, previously tested as developmental enhancers based on sequence constraint but without prior tissue specificity predictions.

Changes made:

1. **Omit needless words**: Removed unnecessary phrases like "the degree of enrichment" and simplified "for each of the three tissues examined here."
2. **Prefer simple words**: Used "based on" instead of "on the basis of."
3. **Use simple subjects**: Focused the subject on "we" to simplify the sentence structure.
4. **Use adjectives frugally**: Limited excessive details, such as "extreme sequence constraint," keeping only what was necessary for clarity.

Scientific Method

i. **Problem/Question:** Develop a question or problem that can be solved through experimentation.
ii. **Observation/Research:** Make observations and research your topic of interest.
iii. **Formulate a Hypothesis:** Predict a possible answer to the problem or question.
iv. **Experiment:** Develop and follow a procedure. Include a detailed materials list. The outcome must be measurable (quantifiable).
v. **Collect and Analyze Results:** Modify the procedure if needed. Confirm the results by retesting. Include tables, graphs, and photographs.
vi. **Conclusion:** Include a statement that accepts or rejects the hypothesis. Make recommendations for further study and possible improvements to the procedure.
vii. **Communicate the Results:** Be prepared to present the project to an audience. Expect questions from the audience.